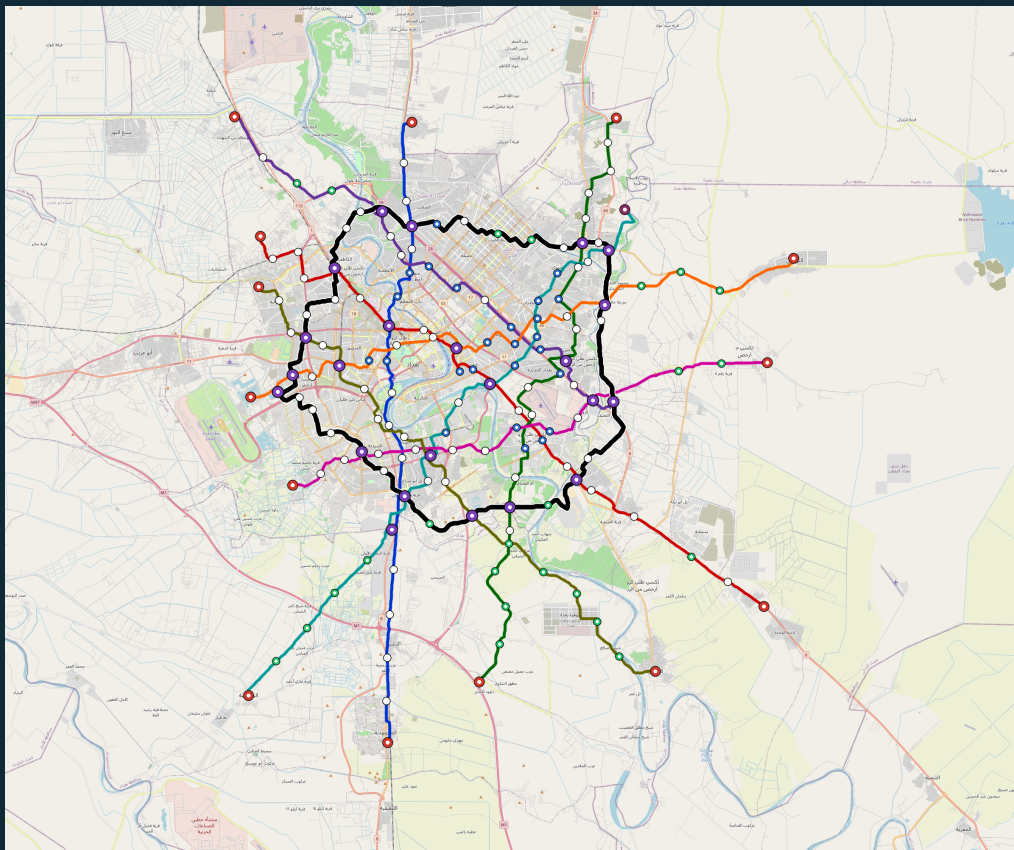


BAGHDAD · IRAQ

# A city-scale railway, designed as a verifiable system

Concept and front-end engineering design offer



**9 lines**

**182 stations**

**516.5 km**

**831 trainsets**

Prepared from source-locked open geospatial data, a deterministic city design, native railway simulation, independent SUMO timetable validation and a city-specific project/operations twin.

STATUS · PLANNING BASELINE — NOT CONSTRUCTION RELEASE · 03 October 2026

EXECUTIVE SUMMARY

# The proposition

OpenSourceRail offers Baghdad an owner-controlled way to move from city data to a buildable railway programme: route planning, topography and water screening, civil classification, fleet and energy sizing, timetable simulation, manufacturing/procurement planning, maintenance, ERP/HR workflows, SCADA boundaries and certification evidence all share one configuration baseline.

|                                                   |                                          |                                                  |                                                |
|---------------------------------------------------|------------------------------------------|--------------------------------------------------|------------------------------------------------|
| <div>516.5 km</div> <div>double-track route</div> | <div>3 min</div> <div>peak headway</div> | <div>2.28 M</div> <div>practical trips/day</div> | <div>\$7.56 bn</div> <div>planning CAPEX</div> |
|---------------------------------------------------|------------------------------------------|--------------------------------------------------|------------------------------------------------|

**Decision requested: authorize a time-boxed owner-led data validation and FEED phase—not construction—then select one priority corridor and a surveyed depot/stabling solution.**

## What Baghdad receives

- A reproducible 9-line network baseline with 182 stations, 23 interchanges and open GIS layers.
- A complete planning twin linking 1,862 assets to budget, procurement, QA, maintenance and operating records.
- A six-car battery-electric fleet concept with candidate CRRC components under competitive procurement.
- Deterministic simulation and standards/FMEA evidence that reruns when controlled design inputs change.
- A staged assurance route that prevents a planning model from being misrepresented as regulatory approval.

## What this document does not claim

It is not a bid from CRRC or any other manufacturer; not a demand forecast, land approval, utility agreement, geotechnical design, issued-for-construction package, safety certificate or financing commitment. Values remain planning screens until the named release gates close.



City-specific QGIS/GDAL package: 9 corridors, 182 stations, 23 interchanges and 2,312 classified civil segments.

PHYSICAL SYSTEM

# Network, terrain and civil concept

|                                         |                                                |                                             |                                            |
|-----------------------------------------|------------------------------------------------|---------------------------------------------|--------------------------------------------|
| <div>428.3 km</div> <div>at grade</div> | <div>75.8 km</div> <div>elevated/viaduct</div> | <div>12.4 km</div> <div>bridge screen</div> | <div>19</div> <div>junction upgrades</div> |
|-----------------------------------------|------------------------------------------------|---------------------------------------------|--------------------------------------------|

The automated designer combines OpenStreetMap transport/building/water/protected-area data with open SRTM elevation and derived slope. It rejects stations predominantly over mapped water and raises likely bridge, terrain viaduct, tight-radius and complex-junction segments for engineering review.

| Civil class        | Segments | Screened length | FEED closure                                                   |
|--------------------|----------|-----------------|----------------------------------------------------------------|
| At grade           | 1112     | 428.3 km        | Survey, land, utilities, drainage, crossings                   |
| Elevated / viaduct | 1143     | 75.8 km         | Vertical alignment, foundations, constructability, urban fit   |
| Bridge             | 57       | 12.4 km         | Hydrology/hydraulics, navigation, scour and authority approval |

## Station and operating layout

The plan includes 182 unique stations and 23 interchange complexes. Stations remain prohibited on mapped-water footprints in the automatic planner. Platform, access, evacuation, universal access, fire/life safety, utilities, traffic management and property interfaces require site-by-site G1 design evidence.

## Depot and stabling status

**OPEN GATE** · The design allocates 831 trainsets operationally, but physical overnight positions, track-by-track capacity, workshop circulation, power/fire separation, security and morning/evening moves are not yet surveyed. The current \$8 M depot allowance is not reconciled and must not be treated as a final depot cost.

Proposed FEED action: compare at least three land-backed depot/distributed-stabling options, freeze an operator-owned fleet deployment plan, then update civil, energy, programme and cost baselines together.

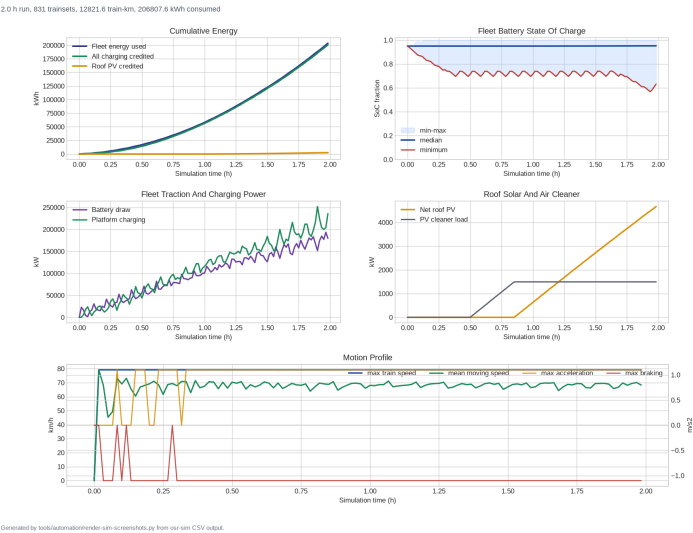
OPERATING CONCEPT

Service, fleet and energy

|                                                 |                                              |                                                   |                                               |
|-------------------------------------------------|----------------------------------------------|---------------------------------------------------|-----------------------------------------------|
| <div>05:30–02:00</div> <div>operating day</div> | <div>751</div> <div>peak revenue fleet</div> | <div>71 + 9</div> <div>spare + cold reserve</div> | <div>3,952</div> <div>one-way trips/day</div> |
|-------------------------------------------------|----------------------------------------------|---------------------------------------------------|-----------------------------------------------|

The full native run covers 90,000 seconds and 216,126 train-km with zero invariant violations. Eight degraded cases—including aged batteries, maximum HVAC load, charging-contact loss, grid outage, missed charges and charger conflict—pass the present software model. SUMO independently records 36/36 planned validation services arriving.

Baghdad simulation trace: motion, dwell, charging, roof PV, and battery state



Native two-hour evidence trace; the full service-day-plus-run-out and degraded cases are retained as hashed JSON evidence.

| Energy element     | Planning baseline | Release boundary                                            |
|--------------------|-------------------|-------------------------------------------------------------|
| Traction demand    | 2,053.8 GWh/year  | Calibrated vehicle/duty data and operator timetable         |
| Station/depot PV   | 52.1 MW           | Surveyed roofs/sites, yield, structure, protection          |
| Stationary storage | 354 MWh           | Supplier product, fire strategy, degradation and duty       |
| Connected charging | 316 MW            | Grid/interface studies, selectivity and timetable conflicts |
| Dedicated solar    | 1,018.6 MW        | Land, connection, PPA/ownership and measured resource       |

The model's zero residual grid/PPA import is a planning result. It is not proof of islanded operability or an approved grid connection; operating energy is explicitly marked unvalidated.

INDUSTRIAL SYSTEM

# Rolling stock with a candidate CRRC parts package

The proposed fleet is 831 OpenSourceRail six-car battery-electric trainsets. The vehicle architecture, requirements, interfaces, FMEA, maintenance identities and acceptance gates remain owner-controlled. CRRC product families are included as candidate components because CRRC publicly lists traction and electrical control equipment and vehicle components—not because a supplier has been appointed.

| Candidate package | Potential CRRC scope                                         | Required acceptance evidence                                                 |
|-------------------|--------------------------------------------------------------|------------------------------------------------------------------------------|
| Traction          | PMSM motors; SiC/traction converters; auxiliary converters   | Duty-cycle sizing, efficiency map, thermal/EMC, HIL and first article        |
| Running gear      | Bogies, air springs, dampers and associated components       | Gauge/axle load, dynamics, fatigue, climate/dust and maintainability         |
| Interfaces        | Couplers/draft gear, brake equipment, train-network controls | ICDs, braking case, cybersecurity, diagnostics, rescue and interoperability  |
| Passenger systems | Passenger-information and equipment interfaces               | Accessibility, language/content, EMC, fire, software and fleet configuration |

## Procurement and localisation controls

- Issue performance-based RFQs and accept multiple compliant examples; do not lock the design to an unqualified catalogue part.
- Freeze actual manufacturer, model, firmware, drawings, material declarations, test reports, warranties, spares and obsolescence plan at G2.
- Qualify local assembly, inspection, tooling, calibration, training and technology-transfer outcomes contractually—not by supplier nationality claims.
- Keep batteries, cooling/HVAC, traction, braking, bogies, doors, train control and rescue interfaces in one system integration baseline.

**Supplier statement · No CRRC partnership, endorsement, quotation, selected product or configuration approval exists in this package. All sourcing is subject to competition, technical qualification, licensing, contract and authority acceptance.**

## Primary public sources

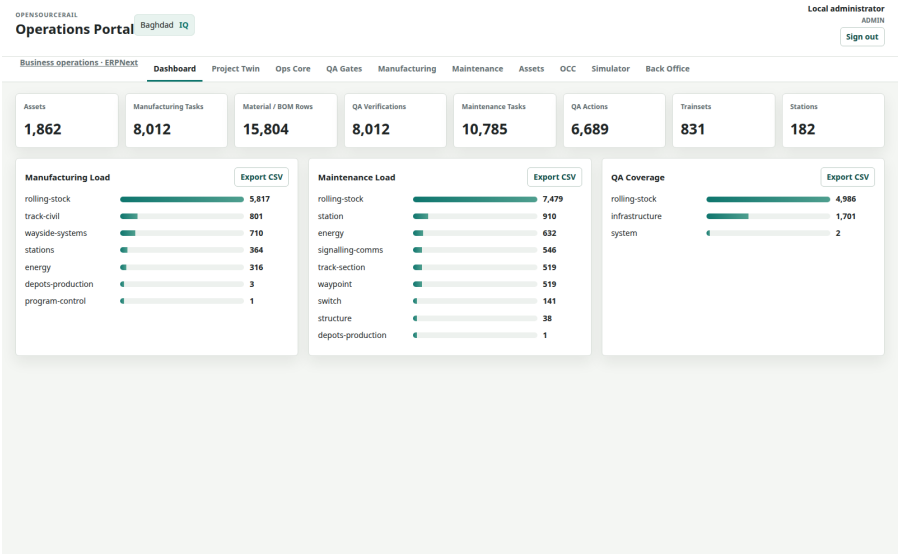
CRRC components overview · vehicle components · electric-control equipment · CRRC Dalian traction/control

PROJECT, ERP, SCADA AND MAINTENANCE

# One digital operating and delivery system

|                                               |                                                 |                                                |                                                |
|-----------------------------------------------|-------------------------------------------------|------------------------------------------------|------------------------------------------------|
| <div>1,862</div> <div>controlled assets</div> | <div>8,012</div> <div>manufacturing tasks</div> | <div>15,804</div> <div>material/BOM rows</div> | <div>10,785</div> <div>maintenance tasks</div> |
|-----------------------------------------------|-------------------------------------------------|------------------------------------------------|------------------------------------------------|

The Baghdad project twin connects engineering configuration to schedule, cash requirements, procurement candidates, QA gates, manufacturing verification, asset registers and preventive maintenance. ERPNext/Frappe remains the business-system boundary; supervision/SCADA observations are isolated from movement authority and protection functions.



Live Baghdad operations dataset rendered by the OpenSourceRail portal; screenshot captured from the local controlled bundle.

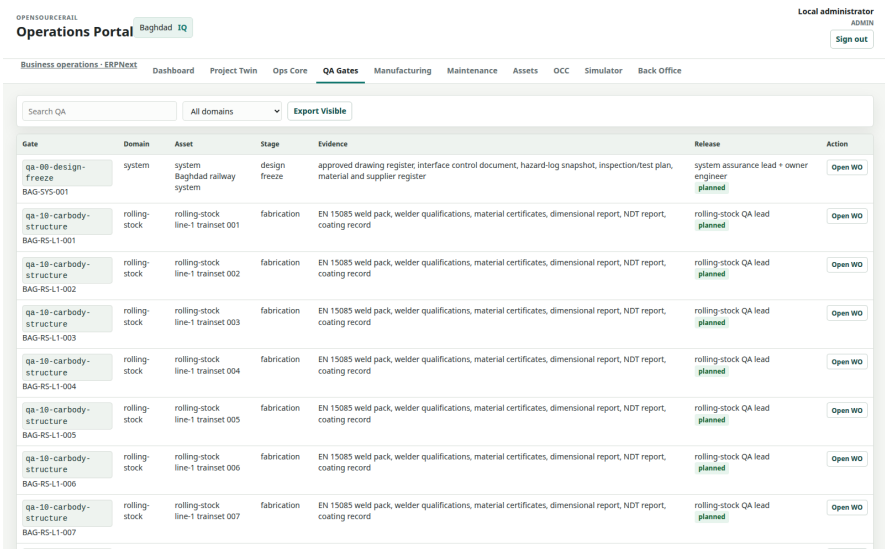
## AI-supported management—with human authority

A multi-model advisory council may draft reversible management actions and compare independent model ballots. Quorum, model diversity, dissent, evidence hashes and an operator attestation are recorded. The council cannot command trains or SCADA, override ATP/interlocking or braking, authorize safety release, hire/dismiss staff, commit money, sign contracts, approve payments or exercise legal authority.

The management model is decision support, not an artificial legal officer. Named competent people remain accountable for every approval and operational authority.

CERTIFICATION AND AUTHORIZATION

# Assurance built into every change



Baghdad QA-gate view: planned evidence is configuration-linked; a planned row is not an approval.

| Gate           | Purpose                      | Minimum decision evidence                                                | Authority                         |
|----------------|------------------------------|--------------------------------------------------------------------------|-----------------------------------|
| G0 · baseline  | Identity and provenance      | Version, source, applicability, checksum and owner                       | Configuration control             |
| G1 · design    | Design assurance             | Requirements, standards, hazards/FMEA, calculations and interfaces       | Owner engineer / discipline leads |
| G2 · qualify   | Actual implementation        | Supplier/product configuration, type/first-article tests and NCR closure | QA + engineering + supplier       |
| G3 · integrate | Installed system             | FAT/SAT, site/vehicle integration, degraded operation and training       | Operator / integrator / assessor  |
| G4 · authorize | Independent/legal acceptance | Independent assessment and competent authority decisions                 | Named legal authorities           |

## Current evidence position

The GIS package, cost/finance generator, native simulation including eight degraded cases, SUMO timetable validation and operations bundle pass their deterministic planning checks and have current hashes. The overall city package remains incomplete by design because two physical summaries—depot scope and stabling—have not passed release readiness.

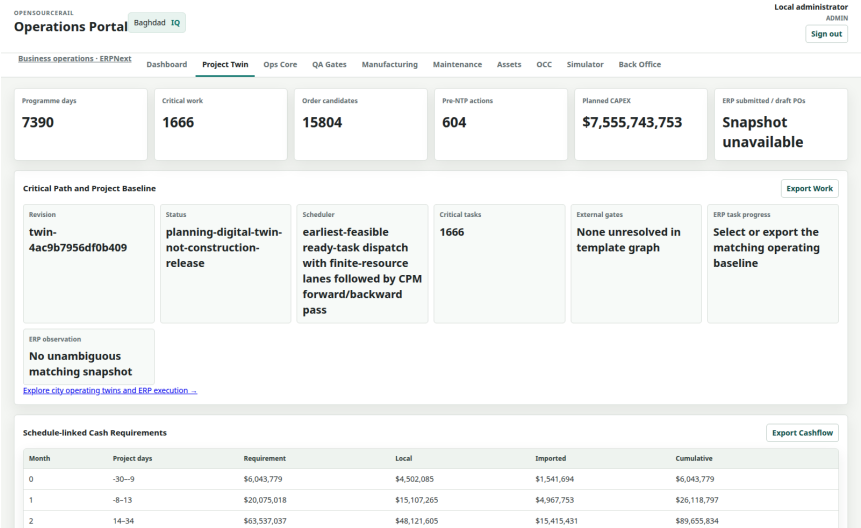
No simulation eliminates required physical testing. The digital twin reduces redesign risk by screening standards, interfaces and failure modes before fabrication, then binds physical evidence back to the exact configuration.

PROGRAMME AND FUNDING

# Commercial planning and delivery route

|                                                               |                                                         |                                                         |                                                              |
|---------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------|
| <div><div>\$7.56 bn</div><div>base planning total</div></div> | <div><div>\$8.69 bn</div><div>+15% envelope</div></div> | <div><div>\$9.44 bn</div><div>+25% envelope</div></div> | <div><div>77.1%</div><div>modelled local capital</div></div> |
|---------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------|

| Planning bucket     | Value     | Critical qualification                                     |
|---------------------|-----------|------------------------------------------------------------|
| Civil works         | \$3.89 bn | Survey/land/utilities/geotechnical and detailed structures |
| Stations            | \$908 M   | Site fit, passenger/fire/accessibility and MEP design      |
| Rolling stock       | \$1.40 bn | Competitive supplier offer, qualification and support      |
| Solar plant         | \$815 M   | Land, grid, measured resource and commercial structure     |
| Depots              | \$8 M     | Not reconciled; replace after depot/stabling FEED          |
| Other systems / EPC | \$534 M   | Control, charging, project services and integration        |



Baghdad project twin: 7,390 planning days, 1,666 critical work packages and 604 pre-NTP actions. Its status is explicitly “planning-digital-twin-not-construction-release”.

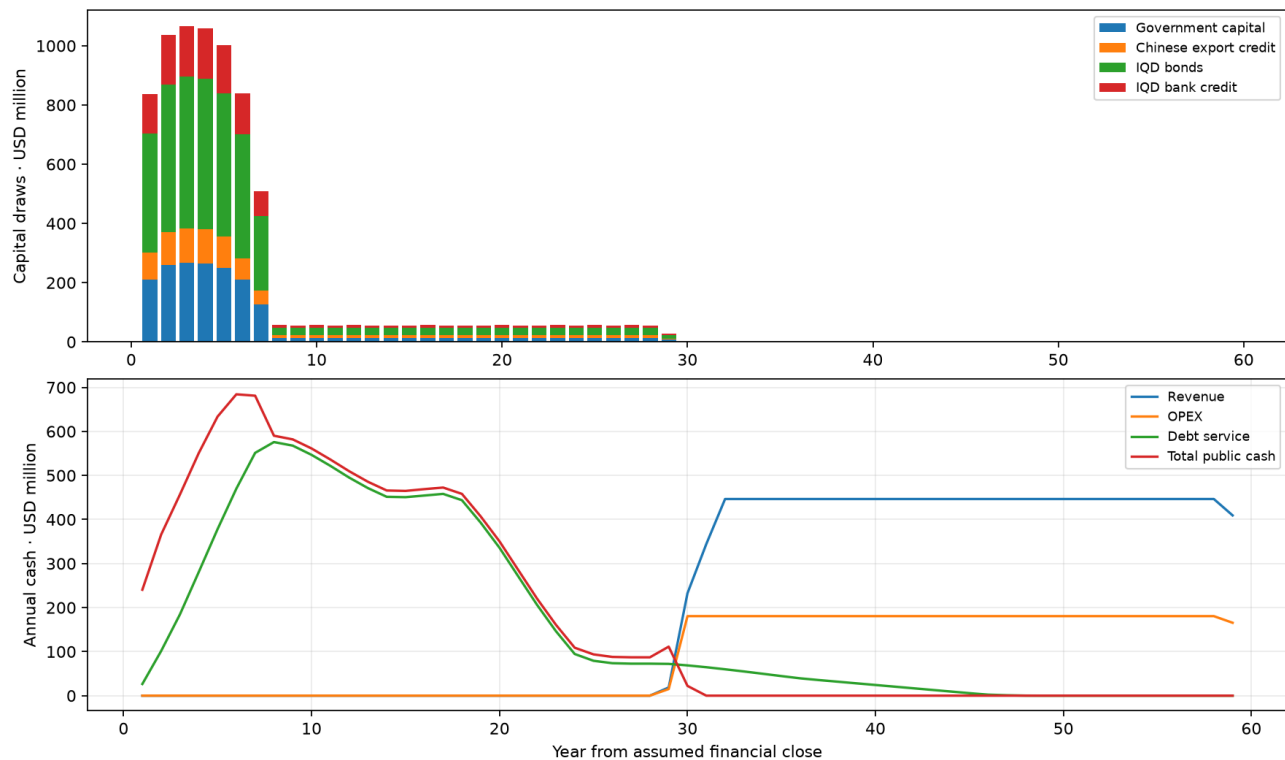
The foreign-turnkey and localisation figures are sensitivities, not received bids. Land, taxes/duties, utility relocation, escalation and unresolved depot scope must be owner-defined before affordability decisions. Financing fees are separately modelled in the Iraq cashflow.

STRUCTURED FINANCING APPRAISAL

# Iraq sources, repayments and public cash

| City capital source    | Currency | Proposed native amount |
|------------------------|----------|------------------------|
| bank credit            | IQD      | IQD 1.561 trillion     |
| Chinese loan           | USD      | \$865 M                |
| domestic bonds         | IQD      | IQD 4.682 trillion     |
| Government import cash | USD      | \$865 M                |
| Government local cash  | IQD      | IQD 1.332 trillion     |

Baghdad · Iraq funding appraisal  
Uncommitted terms; capacity-led low-demand case



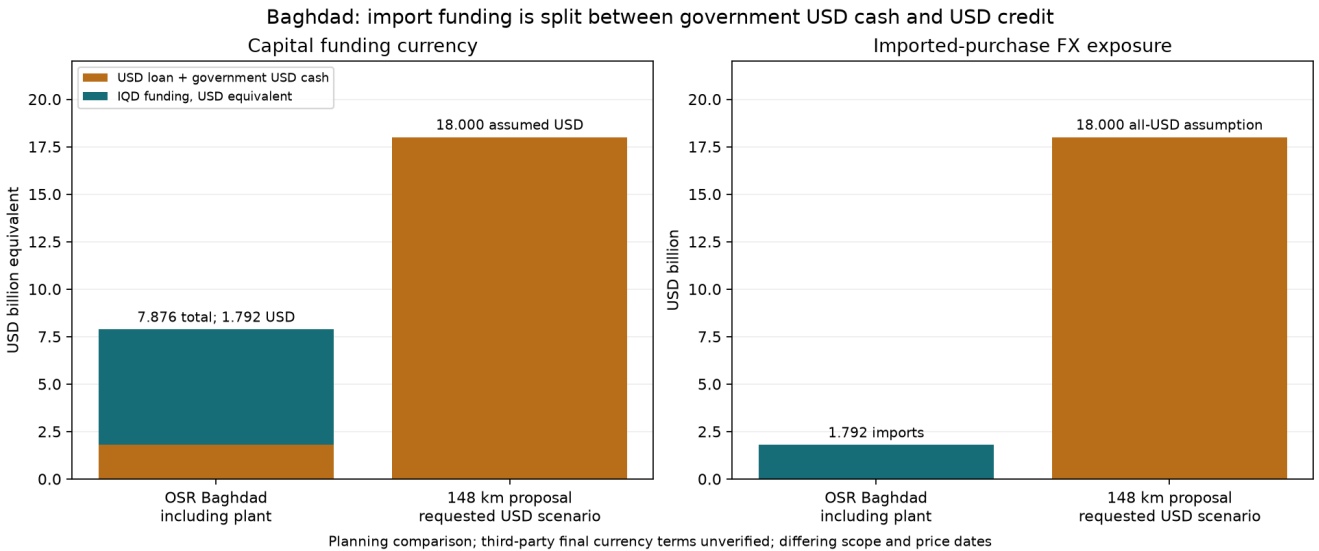
Construction cash spans 347 assumed calendar months under the current constrained CPM. Peak annual conditional public cash requirement is \$684 M; loan repayment is tied to draw dates, not postponed until opening.

The import basket is assumed eligible for proposed Chinese export credit pending supplier-origin and lender qualification. Imported purchases are split 50% government USD cash / 50% Chinese USD loan. Government capital is 25% of total uses, including invoice downpayments. Proposed IQD bonds and term bank credit split the remaining balance after government and Chinese credit 75:25. Tooling is counted once in the Baghdad-only programme outside city CAPEX. Fees, interest, reserves and later cash shortfalls remain additional unfunded requirements if public cash is limited to that contribution. Loan advance, rates, maturities and origination fees are uncommitted assumptions, not bank offers.

Monthly ledgers separately show native-currency draws, principal, interest, fees, fare/nonfare receipts, OPEX, public support and restricted reserves. FX, demand, delay, short-bullet bonds, withheld appropriations and declined Chinese credit are stressed. Subsidy does not increase pre-support DSCR; no automatic refinancing or unlimited bridge credit is assumed.

HISTORICAL PROPOSAL COMPARISON

# Baghdad: USD capital intensity and local industrial value



| Measure                    | OSR plan including plant | Historical / requested scenario |
|----------------------------|--------------------------|---------------------------------|
| Route / stations           | 516.5 km / 182           | 148 km / 64                     |
| Capital USD equivalent     | \$7.88 bn                | \$18 billion reported           |
| USD capital funding        | \$1.79 bn                | \$18 billion assumed USD        |
| Chinese USD loan           | \$896 M                  | Debt/government split unknown   |
| Government import USD cash | \$896 M                  | Unknown final split             |
| Remaining funding          | IQD 7.909 trillion       | No IQD in requested scenario    |

Sources: July 2024 report; NIC DBOMFT notice.

The July 2024 report gives 148 km, seven lines, 64 stations and an estimated \$18 billion. The NIC DBOMFT notice requires a bidder funding plan. Entirely USD foreign loans and government cash are the requested comparison assumption; contracted currency, fares and debt terms have not been verified. Historical and OSR planning budgets differ in scope, price date and maturity; this is not a qualified bid saving.

Only the Chinese loan is USD debt. Half the imports are government USD cash inside its 25% capital contribution. Remaining government cash, bonds, bank credit and local revenue use IQD. Imported-purchase FX exposure is \$1.79 bn; IQD debt matching reduces currency mismatch but does not remove Chinese repayment FX risk or domestic interest and placement needs.

## SUSTAINABILITY APPRAISAL

## Affordable service and direct Iraqi benefits

Modelled average fare: IQD 1,317 per paid trip. Thirty trips use 8% of the retained income proxy; 44 commuter trips use 11.7%. The low case assumes 1,138,320 paid trips/day, not unique residents or calibrated demand. At that volume, OPEX-only neutrality requires IQD 485; this excludes debt and reserves.

First complete operating year cash neutrality including debt and fees requires about IQD 1,482 per trip at fixed ramped demand, excluding factory debt and reserve deposits. This is a threshold, not a recommended fare. The full-network-only case with 347 capital months leaves \$9.76 bn in unfunded additional cash requirements under the public cap; higher fares alone cannot finance that construction-period burden.

Conditional phased opening starts line revenue in month 66 and reaches all lines in month 346. Programme additional liquidity needs fall to \$8.26 bn. Each line has its own revenue ramp; fleet shares proxy demand and variable OPEX, with 25% fixed OPEX from first opening. The new plant gates train production for 520 working days. Actual commissioning remains pending. Later surplus is retained: these gross cash injections are not net lifetime loss.

Population access is unresolved: the 46.4% anchor score applied to 9,780,429 planning residents yields a 4,538,119 resident proxy. It is not a measured 800 m walking catchment. The older reported 80% city-coverage ambition has no comparable access definition. More route kilometres and stations do not prove greater population coverage.

Potential Iraqi procurement is \$6.08 bn, covering civil works, stations, train assembly, body modules, fit-out, wiring, inspection and maintenance. Imported bogies, batteries, windows, doors and tooling still need qualification. Local production can retain skills, supplier income and repair capacity; no GDP multiplier or tax recovery is booked.

The operating allowance is 2,350 indicative FTE and IQD 19.50 billion annual labour cost. These are not manufacturing/construction job counts; validated hours, productivity, wages and local-content contracts are required. The third-party plan may also employ Iraqi civil labour.

Sustainability requires accepted phasing, surveyed demand, an affordable tariff, funded interest/reserves and debt service within the public limit, placed IQD facilities, local supplier/process qualification and lifecycle replacement provision. Software and ledger checks do not close these decisions.

SIX-MONTH FUNDING PROGRAMME

# Reconciled cash needs and financing alternatives

Capital sources and uses remain \$7.88 bn. Independent pooled cash requires \$8.26 bn early and retains \$8.19 bn later: \$67 M net nominal deficit before charging for gap finance. EPC overhead now follows direct works; capital principal repayments are not extra construction CAPEX.

| Existing annual receipts     | IQD billion, full steady operation |
|------------------------------|------------------------------------|
| Passenger tickets            | 547.335                            |
| Station shops / kiosk leases | 12.984                             |
| Advertising space            | 20.175                             |

These receipts already reduce the gap. Retail assumes 88% occupancy and advertising 85%; rates follow the historical income proxy. Each line's receipts ramp with opening. Tenant demand, collection losses and dedicated concession costs remain unqualified; new income must be additional and net of costs.

Six-month placement envelopes specify native government cash, Chinese loan draws, IQD bond face and unit counts, IQD bank draws, first/last repayment dates, reserve movements and supplemental liquidity. Settlement is monthly against expenditure at par; selling each entire envelope upfront would need a new interest/carry calculation.

| Gap-finance sensitivity | Peak IQD tn | Uncovered IQD tn | Unpaid at end IQD tn |
|-------------------------|-------------|------------------|----------------------|
| commercial gap credit   | 13.000      | 39.260           | 13.000               |
| concessional gap credit | 13.000      | 1.493            | 10.644               |
| blended candidate       | 12.920      | 0.000            | 9.400                |

The illustrative IQD facility has a 13 trillion maximum outstanding balance and pays its own interest/fees. The candidate combines eligible green debt replacing conventional bonds, an uncommitted climate grant, net development-rights proceeds and new net receipts. None is a funding commitment; a green label alone changes no cashflow.

With constant nominal fares and OPEX, eliminating both uncovered cash and terminal debt requires about \$134 M/year of genuinely new net receipts under the other blended assumptions. Indexed fares and costs are tested separately on the next page. IQD on-lending or a priced hedge is needed for foreign concessional funding to preserve the currency strategy.

Detailed calculations, full six-month CSVs and primary-source financing routes are in `engineering/finance/FUNDING-RECONCILIATION.md` and the Iraq finance directory.

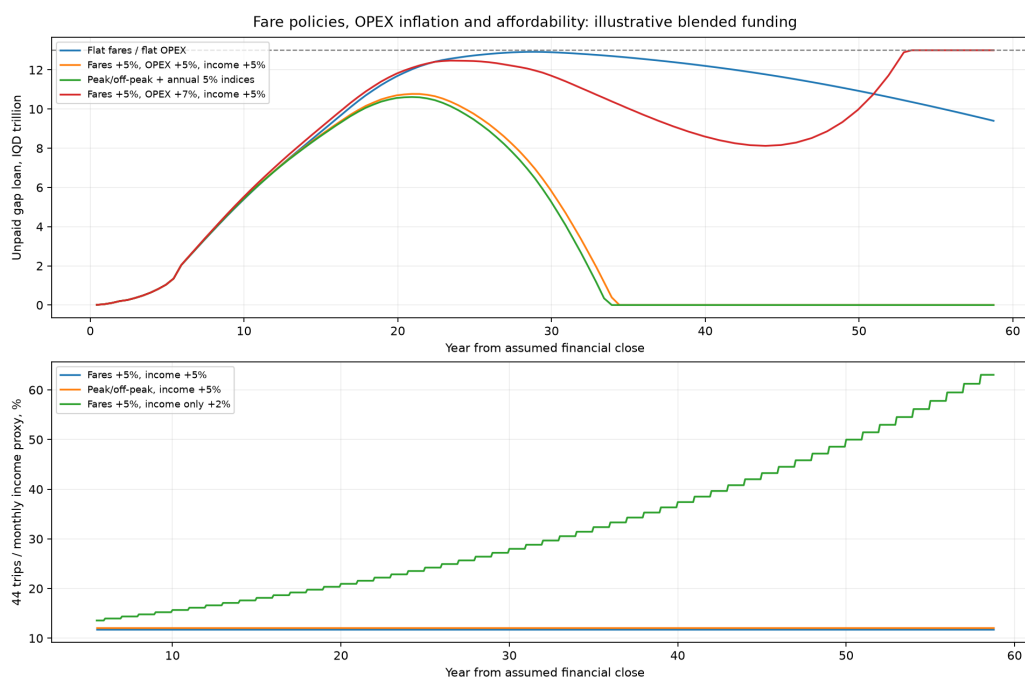
## FARE AND INFLATION SENSITIVITIES

## Variable tickets and indexed operating costs

A 5% annual ticket increase is tested alongside OPEX inflation, income growth and demand response. Annual indices start at financial close; no receipts enter before opening. Peak/off-peak tiers are indicative, with a separate demand response for each tier. No tariff or inflation forecast is adopted.

| Policy                        | Peak debt IQD tn | Uncovered IQD tn | End debt IQD tn |
|-------------------------------|------------------|------------------|-----------------|
| Fares / OPEX / income +5%     | 10.774           | 0.000            | 0.000           |
| Peak/off-peak plus 5% indices | 10.613           | 0.000            | 0.000           |
| Fares/OPEX +5%; income +2%    | 11.734           | 0.000            | 0.000           |
| Fares/income +5%; OPEX +7%    | 13.000           | 11.768           | 13.000          |

In the paired 5% case, first-opening tickets average IQD 1,681; full-opening tickets average IQD 5,164. With matched income growth, 44 trips remain 11.7% of the income proxy. Slower wage growth raises that burden and reduces trips under the assumed -0.30 real-price elasticity.



Uncommitted 2% IQD facility capped at 13tn; cash beyond cap is uncovered; rates and income growth are sensitivities

Later surplus repays the illustrative facility under paired 5% assumptions, while substantial early IQD borrowing, the candidate grant/rights receipts and fixed nominal loan terms are still required. The unlevered NPV before grants/new rights/net-receipt targets is \$-4121 M at 13.4% nominal discount. Distant nominal surplus is not present-value wealth.

Income, occupancy and elasticity are uncalibrated; capital escalation, replacement inflation, FX, variable interest rates and concessional IQD lending need qualification. Affordable commuter/student concessions and transfer caps need explicit compensation rather than assumed free reductions.

## PROPOSED ENGAGEMENT

# A controlled path from model to railway

| Phase               | Indicative purpose                                                                                                                            | Exit decision                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 0 · Validate        | 12–16 weeks: authority map, data room, demand/operations calibration, survey procurement, priority corridor and depot longlist                | Owner accepts requirements, evidence gaps and FEED scope    |
| 1 · FEED            | Surveyed alignment, utilities/property, geotechnical/hydraulic work, depot/stabling options, system requirements, RAMS and cost/risk baseline | G1 design baseline and independently reviewed business case |
| 2 · Procure/pilot   | Competitive component/works packages; supplier qualification; HIL/FAT; one priority corridor, depot and operating organisation                | G2/G3 evidence and restricted trial authority               |
| 3 · Authorize/scale | Independent assessment, trial operation, competence, emergency readiness, legal approvals and measured performance                            | G4 passenger service authority, then controlled rollout     |

## Immediate owner decisions

- Name the legal project owner, transport operator interface, infrastructure authority, safety/regulatory route and independent assessor.
- Authorize a controlled data room and field-data plan; confirm Arabic/English document and operational requirements.
- Select the passenger-demand method, target fares/service policy and a representative priority corridor.
- Protect land for at least three depot/stabling alternatives before choosing a physical configuration.
- Approve performance-based market engagement for rolling stock components, energy, civil works and system integration.

## External gates before any construction release

**Surveyed alignment, property and utilities · calibrated demand and operator timetable · geotechnical, drainage, structural and fire acceptance · supplier-frozen battery, charger, traction and mechanical equipment · independently assessed safety case and competent legal authorization.**

## Evidence and reproducibility

The offer PDF, design, GeoPackage, simulations, operations bundle and screenshots are retained in the Baghdad city package. ``offer/manifest.json`` records SHA-256 hashes for the PDF and its controlled inputs. Regenerate the city with ``./osr city baghdad``, capture the portal with ``node tools/automation/capture-city-offer.mjs``, and rebuild this document with ``python3 tools/automation/build-baghdad-offer.py``.

**OpenSourceRail · open engineering baseline · owner-controlled evidence · fail-closed release claims**